

Dalhousie University
CSCI 3110 — Design and Analysis of Algorithms I
Fall 2024
Assignment 4

Distributed Tuesday, October 1 2024.

Due 11:59PM, Tuesday, October 15 2024.

Instructions:

1. Before starting to work on the assignment, make sure that you have read and understood policies regarding the assignments of this course, especially the policy regarding collaboration. <https://dal.brightspace.com/d2l/le/content/338900/Home?itemIdentifier=D2L.LE.Content.ContentObject.ModuleC0-4457275>
2. To submit via Crowdmark, upload a **separate** image/PDF file (multiple pages are allowed) to each question. You may resubmit as often as necessary until the due date. A good strategy is to create an initial submission days in advance after you solve some problems, and make resubmissions later. More detailed submission information for Crowdmark can be found at:
<https://crowdmark.com/help/completing-and-submitting-an-assignment/>
3. If you submit a joint assignment, use Crowdmark to create a group and add group members before making submissions. The “What Will Students See” section of <https://crowdmark.com/help/creating-a-group-assignment> shows the steps and screenshots of creating a group and adding members. Note that, “to avoid overwriting submissions”, Crowdmark recommends “only one group member submits all the work”.
4. We encourage you to typeset your solutions using LaTeX. However, you are free to use other software or submit scanned handwritten assignments as long as **they are legible**. We have the right to take marks off for illegible answers.

Questions:

1. [10 marks] For each of the following four recurrences, use the “master theorem” and give the solution using big- Θ notation. Explain your reasoning. If the “master theorem” does not apply to a recurrence, show your reasoning, but you need not give a solution.
 - (a) $T(n) = 9T(\lceil n/3 \rceil) + \Theta(n^2 \lg^3 n)$;
 - (b) $T(n) = 4T(n/2) + n^2/\lg n$;
 - (c) $T(n) = 7T(n/3) + \Theta(n^2)$;

(d) $T(n) = 3T(\lfloor n/5 \rfloor) + 3T(\lceil n/5 \rceil) + 200n \lg n.$

2. [10 marks] Now, consider the problem of finding the L_1 -closest red-blue pair defined in Assignment 3 again. This time we consider a different special case. In this case, there is a horizontal line $y = h$ such that all red points are above this line, and all blue points are below this line. Use divide-and-conquer to design an $O(n \lg n)$ -time algorithm for this special case.

You may use the solution to Question 2 of Assignment 3 as a subroutine without describing how it works; even if you do not have a solution to Question 2 of Assignment 3, you can still assume the existence of an $O(n)$ -time algorithm for it and use it in your solution to this question. You may continue to assume that all coordinates are distinct.

3. [10 marks] Give an $O(n \lg^2 n)$ -time algorithm that solves the problem of finding the L_1 -closest red-blue pair in the general case, by using divide-and-conquer. You may use the solution to Question 2 of this assignment as a subroutine, and assume that all coordinates are distinct.